TFPT — Red Team / Stress Test layer

Five adversarial targets (A–E): try to *break* the reductions, not confirm them

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June 28, 2026 · v5.3 (rev 326)

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\mathrm{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order (with **four** companions):

- **0.** `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
- **1.** `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
- **2.** `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
- **3.** `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
- **4.** `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
- **5.** `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\mathrm{geo}}, G_{\mathrm{net}}, F_{\mathrm{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

You are reading document 5 (Red Team).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\rm car} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
$v_{\rm geo}$	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
$G_{\rm net}$	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$	[C]
$F_{\rm transfer}$	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Gravity is parameter-free. The classical metric-sector field equation is no longer only an $R+R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2| 2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status and the one unit $v_{\rm geo}$; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem $G_{\rm net}$ (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) is now *closed modulo a cited theorem.* The explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD = 1 (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual that stays [O] is the abstract continuum existence of the scaling limit, i.e. exactly those two cited published theorems (`v336`); so this is *closed modulo a cited theorem*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76`/`v330`). So the honest residual is the one unit $v_{\rm geo}$ plus the typed $F_{\rm transfer}$ interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels $(U_{\rm wall}, G_{\rm metric}, F_{\rm frontier})$ are retained only for ledger continuity — the live residual is $v_{\rm geo} \oplus F_{\rm transfer}$, with the seam $G_{\rm net}$ closed modulo a cited theorem and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\mathrm{car}} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1,1,2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

↓

Physical bridges: v_{geo}, G_{net}, F_{transfer} *([C] bridges)*

↓

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

What this note is

This is the deliberately *adversarial* layer requested in the v78 review. Its purpose is **not** to confirm TFPT but to attack the five load-bearing reductions at their weakest logical transitions. Each target is run through one fixed protocol — minimal statement, assumptions, logical chain, validity conditions, counterexample search, limiting/degenerate cases, alternative structures, provisional verdict — and produces a deliberately narrow output: where the statement works, where it could fail, and the residual risk. The layer is machine-backed by [`verification/redteam/rt_A_e8net.py`] … [`verification/redteam/rt_E_vgeo.py`], [`verification/redteam/rt_F_qft4d.py`] and [`verification/redteam/run_redteam.py`]; a red-team check asserts an *adversarial* fact (a counterexample really exists, a hidden assumption is really needed, a firewall really holds), and the honest outcome lives in the **status** of each target, never in a green pass.

Target	Minimal claim	Status	Residual risk
A	seam–Calderón measure = $(E8)_1$ net	[C] closed mod. cited thm	*one* statement: boundary-net holomorphy + c=8 ($\Leftrightarrow$ the index-4 inclusion); E_8 and bulk uniqueness then automatic (v83/v87/v89; Lie-level realisation v143). Now closed modulo a cited theorem: the lattice model (v367/v368) and the S_3 stack (c=8, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, Lean-formalised as `FORM.SEAM.MMST.01`; only the cited MMST/OS continuum existence stays [O] (v336). The free-bulk premise is a fixed-point theorem, the cone is eliminated (cone gap = one-particle gap), and the residual factors into the A_2 net assembly + `GATE.QGEO` (v160–v165).
B	$g_{\mathrm{car}} = 5$ forced by the Pascal rule	survives (narrowed)	boundary derivation of half-spinor exhaustion (degree-2 truncation)
C	$k = c_3/2 = \frac{1}{16\pi}$, $S/A = \frac{1}{4}$	survives	absolute $1/G$ (UV-sensitive) is the one anchor
D	ratios+products $\Rightarrow$ one scale v_{geo}	survives (narrowed)	CP phases ($\delta_{\mathrm{CKM/PMNS}}$) not covered by v_{geo}
E	one scale v_{geo} carries the theory	survives (narrowed)	EW/reheating/leptogenesis scales + seam=Planck identification

Method

The red team treats each reduction as a hostile witness. A confirmatory script that always passes is worthless here: the layer is built so that it *may* downgrade a claim. Four outcomes are allowed — **survives** (stands as worded), **survives (narrowed)** (stands only after a silent assumption is made explicit), **reduced, not closed** (the conservative wording is the correct one), and — reached by Target A after the lattice/S_3 round — **closed modulo a cited theorem** (pinned at every computable level, only an external published existence theorem left). A fifth, **broken**, is reserved for an actual failure; none occurred.

Target A — the $(E8)_1$ boundary-net identification

Minimal claim. The ambient metric/projective measure is reduced to identifying the seam–Calderón boundary measure with the $(E8)_1$ lattice net, carrier subnet $(D5)_1 \times (A3)_1$. [`verification/redteam/rt_A_e8net.py`]

Logical chain (established). The Sugawara level-1 central charges are $c(E8)_1 = \frac{248}{31} = 8$, $c(D5)_1 = 5 = g_{\mathrm{car}}$, $c(A3)_1 = 3 = N_{\mathrm{fam}}$, so the conformal-embedding criterion $c_{\mathrm{coset}} = c(E8) - c(D5) - c(A3) = 0$ holds [E]. This is *compatibility*.

Where it breaks: central charge underdetermines the net

Counterexample: $(D8)_1 = SO(16)_1$ also has $c = \frac{120}{15} = 8$, but is *not* holomorphic (four primaries $1, v, s, c$) — a distinct $c = 8$ net. Hence equal central charge is *necessary, not sufficient*: $(E8)_1$ is the unique *holomorphic* $c = 8$ net (even unimodular rank-8 lattice $= E8$), so **holomorphy (single vacuum sector, μ-index 1) is the load-bearing extra assumption**. Dropping chirality is worse: the $c = 8$ Narain family is positive-dimensional. The constructive map seam–Calderón kernel $\to (E8)_1$ and the uniqueness of bulk reconstruction are not established; the gap $\Delta_{\text{eff}} > 0$ *supports* tightness (clustering) but does not prove it.

Verdict. Keep "reduced to a rigorous boundary-net identification problem"; do *not* write "metric sector closed". The red team confirms the conservative wording. Status [C]. Residual: holomorphy proof + constructive boundary map + bulk-reconstruction uniqueness.

Target B — carrier rank / the Pascal condition

Minimal formal claim. $2^g = g^2 + g + 2$ has the unique solution $g = 5$ (Lean-verified [E]). This arithmetic is *not* attacked. The attack is upstream: why must the carrier obey the Pascal half-spinor exhaustion $2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$? [`verification/redteam/rt_B_pascal.py`]

Where it narrows: the rule, not the arithmetic, selects 5

Perturbing the constant breaks it: $2^g = g^2 + g + c$ gives $\{5\}$ only at $c = 2$; neighbours give other or empty solution sets. The truncation *degree* is the real choice: with $2^{g-1} = \sum_{k \le K} \binom{g}{k}$ one finds $K{=}0 \to g{=}1$, $K{=}1 \to 3$, $K{=}2 \to 5$, $K{=}3 \to 7$, i.e. $g = 2K + 1$. So choosing $K = 2$ (to reach $g_{\text{car}} = 5$) is a physical postulate. It is corroborated — the $D5$ half-spinor dim $S^+ = 2^{g-1} = 16 \Leftrightarrow g = 5$ and the family closure $N_{\text{fam}} = (2^{g-1} - 1)/g$, $g + N_{\text{fam}} = 8$ both pick 5 — but each is itself a postulate, not a theorem.

Verdict. Theorem A's arithmetic core stands; the Pascal *selection* must be typed [O]/[C], not [E]. Residual: a boundary/seam derivation of half-spinor exhaustion.

Target C — $k = c_3/2$ and the seam-area coefficient

Minimal claim. In reduced units $k = c_3/2 = \frac{1}{16\pi}$ and Fursaev–Solodukhin gives $S/A = \frac{1}{4}$. [`verification/redteam/rt_C_kc3.py`]

The dimensional firewall holds

The only legal chain is

$$k_{\text{red}} = \frac{c_3}{2} = \frac{1}{16\pi} \ \text{(dimensionless)}, \qquad k_{\text{phys}} = \frac{k_{\text{red}}}{G}, \qquad S = 4\pi\, k_{\text{phys}} A_{\text{phys}} = \frac{A_{\text{phys}}}{4G}.$$

The forbidden form $k_{\text{phys}} = c_3/2$ is dimensionally false (legal only at $G = 1$). A scan of `v67`/`v68`/`v73` finds every $c_3/2$ used as the *reduced* coefficient and **zero** naked dimensionful identities; the firewall's teeth are verified against a synthetic violation. Internally consistent with $1/G$ being UV-sensitive (Sakharov/Connes, v68).

Verdict. Safe as worded (reduced coefficient). Status [E] (structure) + [O] for the residual. Residual: the absolute $4D$ Newton scale $1/G$ ($\propto \Lambda^2 f_2$) stays the one dimensionful anchor; nothing *dimensionless* is open.

Target D — $U_{\text{point}} \to v_{\text{geo}}$

Minimal claim. Ratios plus sector products determine the dimensionless amplitudes, leaving one common scale v_{geo}. [`verification/redteam/rt_D_upoint.py`]

Where it narrows: the bijection needs explicit hypotheses

For positive reals, (ratios, product) $\Leftrightarrow$ (individuals) is a bijection, and the TFPT sectors satisfy it (positive rational c, odd size $3 = N_{\text{fam}}$). But it *fails* off-assumption: even sector size leaves a sign ambiguity ($\{2, 3\}$ vs $\{-2, -3\}$ share ratio 3/2 and product 6); complex amplitudes leave an n-th-root branch ambiguity ($\{1, 1, 1\}, \{\omega, \omega, \omega\}, \{\omega^2, \omega^2, \omega^2\}$ share all ratios and product 1); a zero amplitude is degenerate. **Genuine residual:** the bijection fixes amplitude *magnitudes* only — CP phases $\delta_{\text{CKM}}, \delta_{\text{PMNS}}$ are complex and are *not* folded into v_{geo}.

Verdict. Holds for TFPT once four hypotheses (real, positive, fixed branch, no zeros / odd sector size) are made explicit; they are silent in v75 and must be named. Status [E] (reduction) + [O] (v_{geo}). Residual: the CP-phase sector.

Target E — v_{geo} as the unique dimensional floor

Minimal claim. No pure-number theory can derive an absolute dimensionful scale; all scales are ratios to one unit v_{geo}. [`verification/redteam/rt_E_vgeo.py`]

Single-scale audit: exact for the certified tiers, conditional beyond

The closed compiler + protected IR readouts all reduce to (pure number)$\times v_{\text{geo}}$: $M_{\text{scal}} = c_3^{7/2}\bar{M}_{\text{Pl}}$, $f_a = (c_3/|\mu_4|)^{7/2}\bar{M}_{\text{Pl}}$, masses $= (\pi/\sqrt{2})\, c_f\, \varphi_0^k\, v_{\text{geo}}$, and $1/G$ shares the same anchor (v68/v75). The audit *surfaces* candidates that are not pure reductions: the seam cutoff Λ (only an *identification* seam=Planck collapses it to v_{geo}), the electroweak matching scale, the reheating temperature, and the leptogenesis scale — all frontier [C]/[O] inputs that must not be silently absorbed into v_{geo}.

Verdict. v_{geo} is the unique dimensional floor of the certified tiers (true by dimensional analysis). Full single-scale-ness is conditional on the flagged identifications staying honestly typed. Status [E] (ratios) + [O] (one scale). Residual: explicit reduction (or honest frontier typing) of the EW / reheating / leptogenesis scales + the seam=Planck cutoff.

Target F — the perturbative 4D-QFT + scale round (v269–v275)

Minimal claim. The round published in release 5.2 — the Epstein–Glaser S_{pert} layer (v269/v271/v273), the PMNS Jarlskog CP strength (v270), the absolute ν-scale typing (v272), the scale over-determination (v274) and the `QG.AMB.01` roadmap (v275). [`verification/redteam/rt_F_qft4d.py`]

Two attacks land; the round survives once they are named

(1) The gravity S_{pert} ghost is a truncation artefact. Power-counting renormalizability is *not* unitarity: the local R^2/Weyl^2 *truncation* gives a four-derivative propagator $1/(p^2(p^2 + M^2))$ with a negative-residue massive spin-2 mode (the Stelle ghost). But that is precisely the Seeley–DeWitt *truncation* of the seam KMS form factor: the entire factor $a(p^2){=}e^{p^2/M^2}$ keeps its only pole at $p^2{=}0$ (v304), the spin-2 sector is ghost-free via the Barnes–Rivers decomposition (v370), and the nearest truncation-zero modulus runs to infinity (v380), so *perturbative* graviton unitarity is established and the ghost is exactly the truncation. The matter+gauge S_{pert} is unitary outright, and now closes *to all orders* as a typed Epstein–Glaser/BRST contract (v381, `QFT4D.EG.ALLORDER.01`): dimension-4 power-counting $\Rightarrow$ a finite counterterm space, BRST nilpotency $s^2{=}0$ for $su(3){\times}su(2)$ (Jacobi), and the seam gap $\Rightarrow$ the adiabatic limit, with the imported all-order T_n existence (causal factorisation) and Slavnov–Taylor identity; the *nonperturbative* `QG.AMB.01` is then a [C] *redundancy* (v369; gap-decoupled from every TFPT readout, not a TFPT structural item, v335), a certification object rather than a new gap. **(2) The over-determination is conditional.** v274's Route-2 inverts the $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ prediction, which is itself `LAMBDA.BRANCH.01` [C]; so the 0.11% gravity-vs-cosmology agreement is conditional on the Λ-branch (the gravity route is unconditional). **Already-honest:** J_{PMNS} inherits δ's [C] provenance (assembled μ_6 phase, not an M_ν output) and the ν-scale is [O] with the NO floor $\Sigma m_\nu = 0.0586$ eV as the one kill test.

Verdict. Survives (narrowed): the round stands as typed; the two narrowings above are now folded into v269 and v274. Residual: the perturbative graviton sector is now established *unitary* — the Stelle ghost is a truncation artefact (v304/v370/v380) — and the nonperturbative `QG.AMB.01` is a [C] *redundancy* (gap-decoupled from every TFPT readout, not a TFPT structural item, v335/v369); and the anchor over-determination is conditional on the Λ-branch. On the all-order footing (v381) the SM-non-unification is robust to all orders, and the *optional* carrier-Pati–Salam UV branch is *proton-safe* ([`verification/v385_uvbranch_killtest.py`]): its minimal $SU(4)$ gauge bosons mediate rare LFV ($K_L \to \mu e$), *not* $p \to e^+\pi^0$, so $M_{PS} \sim 3\times10^{13}$ GeV clears the binding bound by $\sim 10^7$ — a frozen kill-test surface (κ/forbidden-rep/new-state/LFV) with *no* fake τ_p window (v265).

Under examination: an infinite-derivative narrowing ([C], does *not* close the gate). The Stelle ghost is a feature of the *local* $R{+}R^2$/Weyl^2 *truncation*: the four-derivative propagator $1/(p^2(p^2{+}M^2)) = (1/M^2)[1/p^2 - 1/(p^2{+}M^2)]$ carries the negative-residue spin-2 mode. But that truncation is the a_4 Seeley–DeWitt order of the spectral action $\mathrm{Tr}\, f(D^2/\Lambda^2)$, whose cutoff is *not* a free scheme — it is the $\beta{=}1$ seam KMS weight $f(u) = e^{-u}$ (v259, no new parameter). Keeping that exponential cutoff *un-truncated* points to a nonlocal, infinite-derivative graviton form factor $p^2a(p^2)$; for an *entire* factor $a(p^2) = e^{p^2/M^2}$ (nowhere zero) the dressed propagator $1/(p^2a)$ keeps its *only* pole at $p^2{=}0$ — no ghost (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006) — whereas any *polynomial* (finite-derivative) truncation has a real zero = the ghost pole. So *if* the resummed form factor is entire, the perturbative ghost is a truncation artefact: a conditional [C] narrowing of this attack. It does *not* close `QG.AMB.01` — the trace regulator f is not yet shown to resum to an entire kinetic form factor $a(\Box)$, and a ghost-free *perturbative* graviton is still not the nonperturbative ambient measure. (`QGAMB.IDG.01`, [C]/[O]; consistent with the `SEAM.EQUIV.01` firewall.) [`verification/v304_idg_nonlocal_ghost.py`]

Sharpened by spin sector ([C]). The Barnes–Rivers spin decomposition makes this precise: of the 5+3+1+1=10 symmetric-tensor components the Stelle ghost is a *spin-2* state, so the entire form factor $a(p^2){=}e^{p^2/M^2}$ cures exactly the spin-2 sector (its only pole stays at $p^2{=}0$), while the wrong-sign *spin-0* conformal mode ($c_{\mathrm{conf}}(4){=}-3/2$) is handled separately by the GHP contour (v334). So the full *perturbative* graviton is ghost-free sector-by-sector under the TFPT-fixed package — still [C] (it inherits the entire-analyticity assumption) and still not the nonperturbative measure. (`GRAV.SPIN2.UNITARITY.01`.) [`verification/v370_grav_spin2_unitarity.py`]

And the "entire" assumption is itself discharged at the truncation-tower level ([E]; [`verification/v380_grav_kms_hessian.py`]). The seam KMS cutoff $f(u){=}e^{-u}$ (v259) gives the kinetic form factor $a(u){=}e^u$ ($u{=}p^2/M^2$), which is entire and nowhere zero — so the spin-2 propagator $e^{-p^2/M^2}/p^2$ has its only pole at $p^2{=}0$. The Seeley–DeWitt expansion is the *Taylor truncation* of a: the $R{+}R^2$ order $1{+}u$ carries the Stelle zero $u{=}-1$, the Weyl^2 order $1{+}u{+}u^2/2$ the pair $-1{\pm}i$, and the modulus of the nearest truncation-zero grows monotonically with the order $(1, 1.41, \ldots, 3.07$ at order $1{\to}8)$ and runs to infinity. So the ghost is *exactly* the truncation and resummation removes it — "entire" means "do not truncate the heat kernel", upgrading the assumption above to a derived statement. (`GRAV.KMS.HESSIAN.01`; the exact off-shell curved-space Hessian identification stays the cited heat-kernel step, and it is perturbative, not the nonperturbative measure.)

Outcome

What the red team changed. No target broke. The hardest gate (A) is now *closed modulo a cited theorem* (the lattice/S_3 round pins it at every computable level, Lean-formalised, only an external published existence theorem left, v367/v368/v376–v379); four targets (B, D, E, F) *survive once a silent assumption is named*; one (C) survives as worded. The net effect is not new physics but sharper typing: the package now states *where* each reduction would fail and *which* assumptions are truly necessary, exactly at the weakest logical transitions. Reproduce with `cd verification/redteam && python run_redteam.py`.

The falsifiability surface is machine-checked ([`verification/v375_observatory_registry.py`]).

The frozen prediction registry (`freeze_file.csv`) is verified by a status-typed CI: every prediction carries a kill criterion (no orphan claims), the headline values re-derive from $\{\varphi_0\}$, and a live scorecard records each one's distance to the current data — θ_{12} vs the JUNO first run (0.3092 ± 0.0087, 0.28σ, compatible), θ_{13} vs NuFIT 6.0 NO (2.0σ, the openly-flagged most-tensioned core prediction), β_{rad} vs ACT DR6 (0.37σ, a hint with systematics pending), and $r < 0.036$ (BK18, compatible). The observatory states its own weakest point rather than hiding it.

Follow-up — the verdicts were then used as a worklist [E] ([`verification/v83_e8net_holomorphic_uniqueness.py`])

Two residuals were attacked directly after the red-team pass; the result is recorded in ledger rows `GATE.METRIC.04` and `CAR.PASCAL.01`.

Target A, residual 1 — closed [E]. At level 1 the number of primaries equals det(Cartan) = |fundamental group|: A_3=4, D_5=4, D_8=4, $\mathbf{E_8}$**=1**. So holomorphy (single vacuum sector, μ-index 1) is *necessary* — it excludes the same-c rival $(D_8)_1 = SO(16)_1$ ($c = 120/15 = 8$ but four primaries $1, v, s, c$) — *and sufficient*: a holomorphic $c = 8$ chiral CFT is the lattice theory of an even unimodular rank-8 lattice, of which there is exactly one, E_8, by the Minkowski–Siegel mass formula (mass $= 1/|W(E_8)| = 1/696729600$, and $|\mathrm{Aut}(E_8)|$ saturates it). The $(D_5)_1 \times (A_3)_1$ embedding $c = 5+3 = 8$ stays *compatibility* (16 primaries vs 1). **Consequence:** the constructive map need only show the seam–Calderón boundary net is holomorphic with $c = 8$ (then E_8 is automatic), so Target A drops from *three* residuals to *two*: (i) boundary-net holomorphy $+\,c = 8$ ($\Delta_{\mathrm{eff}} > 0$ supports, not proves), and (ii) bulk-reconstruction uniqueness — both still [C]/[O].

Target B — reduced [E]. The "$K = 2$ Pascal truncation" is not free: $K = (g-1)/2$ is the Pascal-row midpoint $\sum_{k\le(g-1)/2}\binom{g}{k} = 2^{g-1}$ (odd g), i.e. the half-spinor split; the carrier is the even Clifford half-spinor $\Lambda^{\mathrm{even}}(\mathbb{C}^5) = 1 + 10 + 5 = 16 = \dim S^+$. So B's residual reduces to the single standard input "carrier = half-spinor of Spin(10)" (the Lean arithmetic core stays [E]).

Target B — a backward certificate ([`verification/v219_icosahedral_mckay.py`]). The choice $(|\mathbb{Z}_2|, N_{\mathrm{fam}}, g_{\mathrm{car}}) = (2, 3, 5)$ that B attacks is, *given* the E_8 closure, the icosahedron: by the McKay correspondence E_8 is the exceptional top of the finite-$SU(2)$ tower ($2T\to\hat{E}_6$, $2O\to\hat{E}_7$, $2I\to\hat{E}_8$), the binary icosahedral group $2I$ (order $120 = |R^+(E_8)|$) has irrep degrees $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ equal to the affine-E_8 Kac marks ($\sum = 30 = h(E_8)$), and it contains the $2, 3, 5$ rotation-axis orders. This does *not* break B's verdict — it is a *backward* certificate (it presupposes E_8), so $g_{\mathrm{car}} = 5$ stays the P2 interface [O]; what it removes is the impression that $(2, 3, 5)$ is an arbitrary triple, since no spherical McKay node sits above the icosahedron.

Typed honestly at this stage: A (i)+(ii) (since merged into the single keystone `SEAM.EQUIV.01` and now *closed modulo a cited theorem*, see Target A's final form below); Target D's CP-phase residual (at $\theta_{13} = 0$ the Dirac phase is undefined, so this needs a texture correction); Target E's reheating/leptogenesis scales and the seam=Planck identification. Target C is a dimensional *floor*, not a gap. **No target was promoted.**

Second follow-up round [E] ([`verification/v87_bulk_uniqueness_reduction.py`], [`verification/v88_cp_phase_audit.py`], [`verification/v86_nstar_reheating.py`])

Target A, residual (ii) — conditionally closed [E]: Target A = ONE residual. Bulk-reconstruction uniqueness is *not* independent of residual (i). By the algebraic-QFT classification of full 2D CFTs over a chiral net (Longo–Rehren; Kawahigashi–Longo–Müger; Bischoff–Kawahigashi–Longo–Rehren), the possible bulks are the haploid commutative Q-systems $\simeq$ physical modular invariants of Rep(A). For a *holomorphic* net Rep(A) = Vect, so the bulk pairing is *unique*. The contrast is machine-checked: the same-c rival $SO(16)_1$ (discriminant category $\mathbb{Z}_2\times\mathbb{Z}_2$, μ-index 4) admits *six* modular invariants — diagonal, $s\leftrightarrow c$ swap, both E_8-extension invariants $|\chi_0+\chi_{s/c}|^2$ (the lattice glue $E_8 = D_8 \cup (D_8+s)$, h_s=1 bosonic), and two twisted pairings — so without holomorphy the $c = 8$ bulk is *multiply* ambiguous; with it, trivially unique. Hence the entire Target A reduces to the single theorem "seam–Calderón boundary net is holomorphic with $c = 8$" ([C]/[O]; ledger `GATE.METRIC.05`). The theorem also has an equivalent, more tractable *index* form ([`verification/v89_carrier_index_lemma.py`], ledger `GATE.METRIC.06`): by Kawahigashi–Longo–Müger, $\mu_A = [B{:}A]^2\mu_B$, the carrier inclusion has Jones index $[(E_8)_1 : (D_5)_1\times(A_3)_1] = \sqrt{16} = 4 = |\mu_4|$ — the glue-group order *is* the inclusion index — and the extension is the μ_4 simple-current extension (all three glue sectors are $h = 1$ currents;

$248 = 45{+}15{+}64{+}64{+}60$). Since an index-4 simple-current extension of the carrier has $\mu = 16/4^2 = 1$, **holomorphy follows from the index statement**: Target A $\Leftrightarrow$ "the seam net contains the carrier net with Jones index 4 as its μ_4 simple-current extension" — both ends constructed objects, the index controllable by Longo theory, the gap supporting finiteness. And *which* extension carries no freedom ([`verification/v92_glue_uniqueness.py`], ledger `GATE.METRIC.07`): exhaustive classification of the carrier discriminant form ($\mathbb{Z}_4{\times}\mathbb{Z}_4$, $q{=}(5x^2{+}3y^2)/8$) shows the extension tower is *completely rigid* — exactly two Lagrangian glues (the two chiralities, identified by the sheet $\mathbb{Z}_2$; the same two objects as the E_8-extension invariants above), and exactly one halfway extension, whose induced form $q = \{0, 0, 0, \frac{1}{2}\}$ *is* the D_8 discriminant form: the $SO(16)_1$ rival is the unique intermediate of the tower, not an arbitrary counter-model. Tower: carrier ($\mu{=}16$) $\to$ $SO(16)_1$ ($\mu{=}4$) $\to$ $(E_8)_1$ ($\mu{=}1$, sheet pair) — nothing else exists. Target A is thereby the *bare* index statement.

Target A, final form — the Simple-Current Extension Theorem [E]/[C] ([`verification/v154_simple_current_theorem.py`]). Assembled into one statement: the carrier net $A = (D_5)_1{\otimes}(A_3)_1$ extended by the isotropic glue $L = \langle(1, 1)\rangle$ has $[B : A]{=}|L|{=}4$, $c{=}5{+}3{=}8$, $\mu(B){=}\mu(A)/|L|^2{=}16/16{=}1 \Rightarrow$ holomorphic $\Rightarrow B \cong (E_8)_1$. The *entire* adversarial residual of Target A is now the single seam-side identification "the RP seam–Calderón net *is* A" — the algebra ($B \cong (E_8)_1$) is exact and machine-checked (v89/v92/v125/v143/v148). That residual is now sharply located: the analytic core ("the kernel is quasi-free") *follows* from a free RP bulk — the boundary marginal of a Gaussian bulk is the compression of a contraction (v155), the *same* premise the area law and the EH mechanism use — and the net is *explicitly constructed and verified*: DtN $= |k|$, 16 free Majoranas, currents 248=120+128, holomorphic character $E_4/\eta^8 = j^{1/3}$ with 248 at level 1 (v156). *Final reduction (v160–v165):* the free-bulk premise is recast as a fixed-point theorem (quasi-free $\Rightarrow$ all higher cumulants $\kappa_{2n}{=}0$, v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap $(2/3)^6$ (v161), whose value is forced by μ_4-equivariance (v162); and the remaining 16-dim identification factors into the two *already-open* items — A_2 net existence (an assembly of established net constructions, [C]) and R_1 = `GATE.QGEO` ("seam boundary $= \mathbb{P}^1\backslash\mu_4$") — with the irreducible core $\{\pi, v_{\rm geo}\}$ a theorem (v165). So Target A carries *no new independent open content*: it is a unification, not an unconditional proof. The two structural residuals are then discharged to a theorem (v175): the CAR second-quantisation functor $\Gamma(t){=}\bigoplus_m \Lambda^m(t)$ makes full-cone reflection positivity reduce *for every m* to the one-particle contraction (verified on the complete 2^{16}-dim Fock space), and A_2 is an assembled, verified $(E_8)_1$ certificate; net existence and full-cone RP become **[E]**. Target A's single residual is therefore the seam realisation, the named keystone `SEAM.EQUIV.01` — *the raw RP seam is the holomorphic $(E_8)_1$ net at $\tau{=}i$* — which is now *closed modulo a cited theorem*: the explicit lattice model (v367/v368) and the S_3 stack ($c{=}8$, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level and it is Lean-formalised as `FORM.SEAM.MMST.01`, so the *only* residual that stays **[O]** is the abstract continuum existence of the scaling limit (the cited MMST/OS theorems, v336); its conformal-deck premise `QGEO.SYM.01` is its *downstream* definitional face, Lean-pinned (`FORM.SEAMEQUIV.01`, v308) to named cited steps plus the derived Recovery gap. The sprint-by-sprint reduction (v176–v302) is recorded in the changelog, not replayed here.

Target D — the CP-phase residual quantified [E] (not closed). With the frozen leading reading $\delta = \pi/3 + 3\lambda_C^2 = 68.65°$: pull vs $\gamma_{\rm PDG} = 65.7° \pm 3.0°$ is $+0.98\sigma$ (survives). Audit, *not* promoted: the data central value sits $0.07°$ from the alternative coefficient reading $\pi/3 + 2\lambda_C^2$ ($|\mathbb{Z}_2|$ instead of $N_{\rm fam}$) — a textbook look-elsewhere trap; the registry keeps coefficient 3 (`REG.FREEZE.01`). Decision threshold: the readings differ by $\lambda_C^2 = 2.88°$, i.e. 3σ-distinguishable once $\sigma_\gamma \le 0.96°$. The J-inversion is flagged magnitude-contaminated (source-scale s_{23} vs M_Z). Ledger `FLAV.CP.01`.

Target E — one flagged scale computed [C]. The reheating temperature is no longer a silent input: $M_{\rm scal} = c_3^{7/2}\bar{M}_{\rm Pl}$ **[E]** + standard physics give $T_{\rm reh} = 9.6 \times 10^9$ GeV and $N_\star(0.05/{\rm Mpc}) = 51.4$ [C] (ledger `COSMO.NSTAR.01`); honestly recorded: this slow Higgs-channel point is *A_s-disfavoured* (-11.4σ; the measured A_s requires near-instantaneous reheating), so the point is conditional on the decay channel and the frozen band $[50, 60]$ stays the surface of record; the leptogenesis scale and the seam=Planck identification remain typed inputs. **Again: no target was promoted; one residual was *merged* (A), one was *quantified* (D), one input was *computed* (E).**

Target D — the CP residual geometrically *located* (not closed) [E]/[C]

The follow-up round only *quantified* the CP residual; this round *locates* it geometrically, in three consistent readings, without promoting it. **(i) A channel typing** ([`verification/v227_degree_exponent_channel_split.py`]). The E_8 split $248 = 120 + 128$ is a magnitude/phase split: the exponents sum to $120 = |R^+(E_8)|$ (the *magnitude* channel, where the $(\text{ratios}, \text{product}) \Rightarrow v_{\text{geo}}$ bijection lives), the fundamental degrees sum to $128 = \operatorname{rank} E_8 \cdot \dim S^+$ (the *phase/glue* channel). The un-covered CP phases of Target D are exactly the 128 phase channel — a clean home, not a new gate; whether 128 also hosts a dark sector is Frontier, *not* asserted (this replaces the over-strong "S^- is dark matter" reading, now sharpened by v430: the other side S^- is matched $\{2, 128\}$ structure, forced-disjoint from the unmapped $\{12, 14, 18, 20, 24\}$). **(ii) A hexagonal modulus** ([`verification/v220_cp_hexagonal_modulus.py`]). The seam deck is the *square* modulus (j=1728, $\mathbb{Z}/4$ clock; the kill-test selects it), which is real and carries no phase; its exceptional partner is the *hexagonal* modulus j=0 (τ=ρ=$e^{i\pi/3}$, $\mathbb{Z}/6$, CM by $\mathbb{Z}[\omega]$). The frozen leading reading $\delta = \pi/3 + 3\lambda_C^2 = 68.65°$ has leading term $\pi/3 = \arg\rho$, the hexagonal CM phase; CP lives in the $\mathbb{Z}/6$ phase *fiber* over the $\mathbb{Z}/4$ seam, not in the deck. **(iii) An orientation** ([`verification/v225_dual_normal_frame.py`]). The dual normal frame $d = a^\top R^{-1} = -\frac{1}{2}(1, 1, -2)$ (Nariai/horizon normal) and $n = (5, -9, 6)$ (first-generation selector, $n^\top R = (8, 0, 0)$) has oriented volume $\det(\mathbf{1}, d, n) = 21 = N_{\text{fam}} \cdot 7$, and rotating by the hexagonal unit gives $\operatorname{Im}\det(\mathbf{1}, d, \rho n) = 21\sin(\pi/3) \neq 0$ — CP sits precisely in the *orientation* of the normal frame, exactly where the magnitude bijection fails. All three are [E] geometry; the physical CP *derivation* stays the Target-D [C] residual (the seam deck is not re-selected, the registry δ is untouched). **(iv) One hexagonal unit, two sheets** ([`verification/v231_cp_mu6_phases.py`]). The reduction is sharper still: *both* CP phases are μ_6 powers of the single hexagonal CM unit $\rho = e^{i\pi/3}$ — $\delta_{\text{CKM}}^{\text{lead}} = \arg(\rho^1) = \pi/3$ (quark) and $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$ (lepton, the neutrino phase lattice) — with $\rho^4 = -\rho$, so $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ is the quark phase times the $\mathbb{Z}_2$ sheet ($\rho^3 = -1$); the frame orientations of (iii) are correspondingly sheet-flipped ($\pm 21\sin(\pi/3)$). So Target D's CP sector is *not* two independent phase inputs but *one* hexagonal unit read on the two sheets — the magnitude residual $3\lambda_C^2$ (quark, v88) and the physical derivation stay [C], but one of the two phase inputs is removed. **(v) The universal triality phase** ([`verification/v233_cp_triality_phase.py`]). In the canonical $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$ factorisation, $\rho = \omega^2(-1)$ with $\omega = e^{2\pi i/3}$ the $\mathbb{Z}_3$ triality centre (the 2/3 cusp weight); since $\rho^4 = \rho^1\rho^3$ with $\rho^3 \in \mu_2$, the quark (ρ^1) and lepton (ρ^4) phases have the *same* family/triality class and differ *only* by the sheet. So the CP phase *is* the universal triality phase, sheet-split — not even a free choice of μ_6 power: the power choice of (iv) is removed, and ω is the μ_3 factor of the order-30 $(2, 3, 5)$ Coxeter/Milnor monodromy that builds the hull (v232). The physical CP derivation still stays [C]. **(vi) The lock as a pre-registered kill test** ([`verification/v320_galois_cp_relation.py`]). Because $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ is now *forced* (one unit $\rho = \zeta_6$, the μ_4-power ρ^4, the $\mathbb{Z}_2$ sheet $\rho^3 = -1$), it is a falsifiable cross-prediction, not a free input: a measured δ_{PMNS} robustly away from 240° at DUNE/Hyper-K (beyond the sub-leading budget, $> 3\sigma$) kills the Galois-CP organisation. The relation is [E], the physical phase reading stays [C], the kill test is [X].

Global look-elsewhere closure [E] ([`verification/v100_numerology_null_mc.py`])

The per-target look-elsewhere traps recorded above (the $\pi/3 + 2\lambda_C^2$ alternative in Target D, the ln(46080) scale reading destroyed in [`verification/v99_koide_flow_time.py`]) are local instances of the global adversarial question: *could the whole scorecard be formula-fishing?* The numerology null test answers it with an explicit, declared null model run in adversarial mode: the *entire* complexity-matched formula grammar (which provably contains every scored TFPT formula — the null space includes the theory) is enumerated against conservative data windows. Joint formula-fishing probability $\prod_i p_i = 10^{-25.8}$ over the 13 scored observables; all 94 500 $F_{U(1)}$ variants solved, exactly one hits CODATA ($p_\alpha \leq 1.1{\cdot}10^{-5}$); total $\leq 10^{-30.7}$. The test is falsifiable in both directions: the negative controls (seed perturbation 1%/10% collapses TFPT's own hit count 13 → 9/6; data shuffling → 1.2) prove it has power, and the census is stable under budget/window variation (< 2 orders). Honest limit, stated as such everywhere: this is a null-model rejection *conditional on the declared grammar* — no statistical test certifies a theory; the tension observables keep their large windows and contribute nothing to the number. Details and the grammar definition: the audit note (`tfpt_3`) and the module docstring.

Structural firewall + out-of-sample seed cross-validation [E] ([`verification/v305_witness_independence.py`], [`verification/v306_seed_crossval.py`])

The global null test answers "is the *scorecard* fishing?"; two companions answer the deeper adversarial worries "is the same integer counted many times as if independent?" and "is one shared seed dressed up as many predictions?". **Witness economy** ([`verification/v305_witness_independence.py`]): every headline integer ($|\mu_4|$=4, g_{car}=5, $|\mathbb{Z}_2|$=2, N_{fam}=3, $\dim S^+$=16, $\operatorname{rank} E_8$=8, 240, 248, $|2I|$=120, h=30,

$|W(D_5)|$=1920, Ω_{adm}=48, the gap exponent 6) is a documented image of the *single* anchor a=(1, 1, 2) with π the only transcendental primitive, so the recurring 2/3=$|\mathbb{Z}_2|/N_{\text{fam}}$ and $(2/3)^6$ are *one* ratio, not many — 13 atoms from 2 free primitives (6.5× compression made explicit). The E_8 spine is genuinely overdetermined (≥ 3 structurally distinct derivations sharing only leaf inputs), while the pure-seed readouts are single-witness, so their joint surprise is scored jointly by v100, never multiplied; the negative control shows α^{-1}'s "137" is *foreign* to the anchor family (an independent witness via the $F_{U(1)}$ cubic, v3). **Seed cross-validation** ([`verification/v306_seed_crossval.py`]): treating the axiom seed φ_0 adversarially as a free dial and back-solving it from each of the five seed-grammar observables gives an error-weighted mean reproducing $\varphi_0 = 1/(6\pi) + 48c_3^4$ to 0.04%; leave-one-out predicts every held-out observable within 3σ *out of sample* (the most-tensioned hold-out is $\sin^2\theta_{13}$, $\sim 2\sigma$), and a data↔formula shuffle explodes the cross-validation χ^2 by > 100× (the test has power). Both stay [E] as structural/statistical checks, conditional on the same declared grammar. **No-golden-dynamics rule** ([`verification/v305_witness_independence.py`]): the golden ratio $\phi = 2\cos(\pi/5)$ lives in the *static* carrier field $\mathbb{Q}(\sqrt{5})$ (the $\mathbb{Z}/5$ hand; minimal polynomial $x^2 - x - 1$, discriminant 5 non-square), while every *dynamic* rate is rational ($\mathbb{Q}$, the $\mathbb{Z}/6$ = sheet×family hand, e.g. the recovery factor $(2/3)^6$) — so a dynamic rate "explained" by ϕ is a cross-field false friend, and CP phases and recovery rates must run on the order-6 hand (the field firewall of v314/v315).

The *reverse* numerology audit, and what the golden ratio means [E]/[O] ([`verification/v354_e8_reverse_audit.py`])

The witness-economy box answers "every readout maps onto E_8 — is the *forward* map fishing?". The deeper adversarial question runs the *other* way: E_8 is a rich object, so how much of its structure maps onto *nothing*, and could the theory be quietly mining that richness? **The reverse audit.** Of E_8's eight Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ (exponents +1), exactly *three* feed a *primary* physical-constant readout — degree 2 (the quadratic/metric), degree 8 (the rank $\to c_3 = 1/(8\pi)$), degree 30 (the Coxeter number $\to g_{\text{car}} = 5$ and the order-30 cycle) — and the other *five* carry no primary readout: $\{12, 14, 18, 20, 24\}$. (They do *coincide* with internal structural atoms — A_3 roots, $\dim G_2$, $\det L$, $|W(A_3)|$, the anchor power sum — as v66 records; but those per-degree coincidences are mostly *unforced*, the numerology temptation that the bandwidth search v355 declines, see the next box.) So 3/8 are primary readouts, 5/8 hull overhead. **Economy, not promiscuity.** A *small fixed* set of E_8 data (rank 8, h=30, $\det K$=1, the $D_5 \oplus A_3$ sub, the 16 half-spinor) generates the *entire* readout set (gauge group, three families, hypercharges, α^{-1}, flavour, scales); the higher Casimirs are never reached into to fit. A numerological use of E_8 would be *promiscuous* (mine the rich structure for matches); TFPT is *economical* (few invariants, many readouts) and never *reaches into* the higher Casimirs to fit; where those non-readout degrees do carry content it is *forced* internal structure (the spine and the flavour det-ladder, v431), not a mined coincidence — the anti-numerology signature is precisely this economy with a forced, *non-readout* remainder. **Why the golden ratio is not numerology.** $\phi = 2\cos(\pi/5)$ sits in exactly that unmapped part: it appears only in *internal* structure (the affine-E_8 attractor eigenvalue v312, the icosian lattice v348) and in no *frozen physical-constant* readout — the sole, *conditional* exception being the axion misalignment spine angle, where $\theta_i = 3\pi/5$ is the regular-pentagon interior angle and $\cos\theta_i = -1/(2\phi)$ ([`verification/v429_axion_pentagon_phi.py`]; [C], branch-level, not a frozen prediction). Numerology means a number *fitted to an observation*; ϕ is fitted to nothing — it is the algebraic signature of the 5-fold (h=30=2 · 3 · 5), the carrier *announcing* that it is icosahedral, and even its one physical-ish appearance is forced geometry (θ_i = the g_{car}-gon angle), not a match to a measured constant. **Honest residual.** The 5/8 hull overhead is exactly that: E_8 is the minimal even-unimodular *hull* (consistency container, $\det K$=1), not the world group, so the higher Casimirs are by-products of unimodular closure — where a future higher-invariant readout could live, or genuine over-capacity (now sharpened: the five non-readout degrees are *not* diffuse slack but the forced two-family ladder 6 · spine ⊔ det-ladder, v431). Stated plainly, not hidden.

Bandwidth search of the unmapped region: the disciplined decline [E]/[C]/[O] ([`verification/v355_e8_unmapped_bandwidth.py`])

The obvious next move — "then *search* those five degrees for hits we overlooked" — is itself the trap: a structure as rich as E_8 returns a numerical match for almost anything. So the scan is run with a *strict discriminator*: a candidate counts only if it is a *forced* identity (a theorem about E_8), never a coincidence. The honest yield is mostly a *decline*, and that decline is the result. **Forced, set-level (the genuinely overlooked structure — collective, not per-number).** $\sum \deg = 128 = \dim S^+$, the spinor half of $248 = 120 + 128$ (a theorem: $\sum \deg = \#\text{positive roots} + \text{rank} = 120 + 8$) — the invariant "budget" of E_8 *equals the fermion content the carrier reads*; $\sum \exp = 120 = \#\text{positive roots}$; $\prod \deg = |W(E_8)| = 696\,729\,600$; and the exponents $\{1, 7, 11, 13, 17, 19, 23, 29\}$ are exactly the $\varphi(30) = 8$ totatives of 30 (v223), so the unmapped degrees' exponents $\{11, 13, 17, 19, 23\}$ are part of that forced set, not free. **Suggestive but *not* a theorem (the discriminator at work).** $12 = h(E_6)$, $18 = h(E_7)$, $30 = h(E_8)$ are all degrees of E_8 (the $2T/2O/2I$ McKay ladder, v219); but "$\deg(E_8)$ contains its subalgebras' Coxeter numbers" is not a general theorem, so this appealing pattern is flagged [C], not claimed. **Declined — the numerology temptation.** The per-degree atom matches $14{=}\dim G_2$, $20{=}\det L$, $24{=}|W(A_3)|{=}4!$, $12{=}|R(A_3)|$, $18{=}p_4(a)$ (which v66 already flagged as fittable for $12, 18$) each admit ≥ 2 readings from the frozen atoms with no forced selector, and the lone extra prime in $|W(E_8)| = 2^{14}3^5 5^2\, 7$ "=" the scalaron exponent is one of many readings of 7 — all declined. **Reconciliation.** v66 (the eight degrees coincide with the load-bearing integers, expected since E_8 is the hull) and the reverse audit (v354: only $2, 8, 30$ are primary readouts) are *both* right once forced set/subchain structure is separated from unforced per-degree coincidence. The search finds *no new forced physical hit*; the disciplined decline *is* the anti-numerology outcome.

The "other side" of the seam vs the unmapped region [E]/[O] ([verification/v430_other_side_reverse_audit.py])

A sharp follow-up runs the audit through the *double cover*: the seam has an *other side* — the one-sided $\mathbb{Z}_2$ collar and the conjugate half-spinor S^- — so *is the leftover where the second sheet lives?* The answer is a disciplined **negative**, the sheet/deck complement of v354/v355. **The deck is the degree-2 invariant (matched).** The one-sided cover contributes $|\mathbb{Z}_2| = 2$ (the $\frac{1}{2}$ in $c_3 = 1/(|\mathbb{Z}_2|\, 4\pi) = 1/(8\pi)$, v58/v216), and $2 = \min\deg(E_8)$ is the quadratic/metric — one of the *three matched* primary readouts $\{2, 8, 30\}$, never one of the unmapped $\{12, 14, 18, 20, 24\}$. **The two sheets are the 128-spinor (matched, collective).** $\dim S^+ = \dim S^- = 16$ and the spinor block is $128 = \operatorname{rank} E_8 \cdot \dim S^+ = 2^{\operatorname{rank}-1} = \sum \deg$, the spinor half of $248 = 120 + 128$; the other side S^- enters 248 as the conjugate $(\overline{16}, \overline{4})$ $(128 = 2 \cdot 64)$, *not* a spare singlet. **Forced-disjoint.** The sheet/deck integer set $\{|\mathbb{Z}_2|, \dim S^+, \dim(S^+{\oplus}S^-), 128\} = \{2, 16, 32, 128\}$ has *empty* intersection with the unmapped degrees $\{12, 14, 18, 20, 24\}$; its only contact with the degree alphabet is the matched 2 (and the collective budget $128 = \sum \deg$). **Disciplined decline [O].** Each unmapped degree, "explained" from the sheet atoms, needs an *unforced* coefficient and admits ≥ 2 readings — exactly the mining the v355 discriminator declines, so there is *no* forced sheet→unmapped-degree identity. **Consequence.** This consolidates the S^--dark-matter downgrade (v227) and the no-spare-singlet WIMP no-go of the frontier E_8 scan: the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live "other side" question — whether the 128 phase/glue channel hosts a dark sector (v227) — concerns *matched* structure, not the unmapped five.

The unmapped five are forced structure, not diffuse overhead: the degree ladder [E]/[C] ([verification/v431_e8_degree_ladder.py])

The reverse audit (v354/v355) and its sheet/deck complement (v430) leave the five degrees $\{12, 14, 18, 20, 24\}$ with no *primary* readout. The structural follow-up asks the sharper question — *are they diffuse hull overhead, or forced internal structure?* — and the answer is the latter, an *exact* two-family decomposition $\deg(E_8) = 6 \cdot \text{spine}\{2, 3, 4, 5\} \sqcup \big(\{2\} \cup \text{det-ladder}\{8, 14, 20\}\big)$. **The split is forced by 30.** The E_8 exponents are exactly the $\varphi(30) = 8$ totatives of $h(E_8) = 30 = 2{\cdot}3{\cdot}5$ (v223), hence coprime to 6, hence $\equiv \pm 1 \bmod 6$ — so the degrees ($= \exp + 1$) occupy *only* the residue classes $\{0, 2\} \bmod 6$: two arithmetic progressions, forced by the prime content of $30 = 2N_{\text{fam}}g_{\text{car}}$. **The $6k$ family is $6\times$ the spine.** Degrees $\equiv 0 \bmod 6 = \{12, 18, 24, 30\}$, divided by 6, give $\{2, 3, 4, 5\}$ — the v91 spine $\{e_3, p_0, e_1, e_2\}$ of the anchor $a = (1, 1, 2)$; $30 = 6g_{\text{car}}$ is the matched Coxeter readout, $12, 18, 24$ were "unmapped." **The $6k{+}2$ family is the quadratic Casimir plus the flavour det-ladder.** Degrees $\equiv 2 \bmod 6 = \{2, 8, 14, 20\}$; here $\{8, 14, 20\} = (\det R, \det C, \det L)$ is the winding line $\det M(s, 0) = 6s + 8$ of the flavour diamond (v135), with $C = R + Q\operatorname{diag}(1, 0, 0)$, and 2 is the matched metric Casimir ($8 = \det R$ is over-determined: also $\operatorname{rank} E_8$). **18 is not a holdout.** $18 = 6{\cdot}3 = 6N_{\text{fam}}$ is the spine-family member $6p_0$, not on the det-ladder ($6s + 8 = 18$ has no integer s): it changes *family*, not status, so *no* genuinely orphan degree remains. **Special to E_8.** The clean $\{0, 2\}$-mod-6 split needs both $6 \mid h$ *and* exponents=totatives(h); among $\{A_4, D_5, E_6, E_7, E_8, F_4, G_2\}$ it holds for E_8 but *fails* for E_6, E_7, D_5, so the decomposition is not a generic mod-6 triviality. **The honest v355 line.** The *arithmetic* decomposition is exact [E]; but the *functorial* claim "the $6k{+}2$ Casimir stratum *is* the flavour sheet operators" needs a representation-theoretic map and stays [C] (degrees $12, 24$ still admit two canonical readings each: 6·spine vs $|R(A_3)|/|W(A_3)|$). So this is a *structure* theorem — it reclassifies the $5/8$ overhead of v354 to *zero* unfamilied degrees — not a forced functorial flavour map, and it closes no gate.

The two structural integers and all of E_8 are fixed by the degree multiset [E] ([verification/v437_e8_degree_joint.py])

A final consolidation of the degree audit (of v6/v66/v355/v431), staying strictly inside the v354/v355 discipline — it adds *no* new per-degree coincidence, only one joint lock. **All of E_8 is fixed by its degrees.** From $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$ alone: rank $= \#\deg = 8$, $h = \max\deg = 30$, #positive roots $= \sum\exp = 120$, #roots $= 240$, $\dim E_8 = 240 + 8 = 248$, $|W(E_8)| = \prod\deg = 696\,729\,600$, $\sum\deg = 128 = \dim S^+$ — every E_8 integer the compiler uses is a function of the degrees. **The joint quadratic.** The two structural integers $(g_{\rm car}, N_{\rm fam})$ are the *unique* root pair of $x^2 - (\operatorname{rank} E_8)\, x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)$: the sum of roots is the rank-fill $\operatorname{rank} E_8 = 8$ (v6) and the product is $h/2 = 15 = g_{\rm car} N_{\rm fam}$ ($2g_{\rm car}N_{\rm fam} = h$, v431), and $\{3, 5\}$ is the unique positive-integer factor pair of 15 summing to 8 — so $g_{\rm car} = 5$ and $N_{\rm fam} = 3$ are forced *together* by two E_8 degree-invariants, not one at a time. The three primary-readout degrees $\{2, 8, 30\}$ are exactly $\{\min\deg, \operatorname{rank}, \max\deg\}$, the invariants every simple Lie algebra distinguishes. Mirrored in Wolfram; closes no gate.

Table 1: The eight E_8 Casimir invariant degrees, audited ([verification/v66_e8_casimir_degrees.py], [verification/v354_e8_reverse_audit.py], [verification/v355_e8_unmapped_bandwidth.py]; the sheet/deck complement [verification/v430_other_side_reverse_audit.py]; the forced two-family structure [verification/v431_e8_degree_ladder.py]). Only three feed a *primary* physical-constant readout; the other five carry no primary readout but are *not* diffuse overhead — they are the forced two families $6\cdot\text{spine}\{2, 3, 4, 5\}$ and the det-ladder $\{8, 14, 20\}$, the residue classes $\{0, 2\} \bmod 6$ forced by $h = 30$ (v431). The genuinely forced collective content: $\sum\deg = 128 = \dim S^+$, $\prod\deg = |W(E_8)|$, exponents = totatives of 30.

deg	exp	TFPT reading	forced?	primary readout?
2	1	$\lvert\mathbb{Z}_2\rvert$	forced	yes: metric
8	7	$\operatorname{rank} E_8$	forced	yes: c_3
12	11	$h(E_6) = \lvert R(A_3)\rvert$	suggestive	no; $6{\cdot}2$ spine
14	13	$\dim G_2$	declined	no; $\det C$ ladder
18	17	$h(E_7) = p_4(a)$	suggestive	no; $6{\cdot}3$ spine
20	19	$\det L$	declined	no; $\det L$ ladder
24	23	$\lvert W(A_3)\rvert = 4!$	declined	no; $6{\cdot}4$ spine
30	29	$h(E_8) = 2{\cdot}3{\cdot}5$	forced	yes: $g_{\rm car}$

collective (forced): $\sum\deg = 128 = \dim S^+$, $\prod\deg = |W(E_8)|$, exp = totatives of 30
two families (forced, v431): $\{0, 2\} \bmod 6$; $6k/6 = \text{spine } \{2, 3, 4, 5\}$, $6k{+}2 = \{2\} \cup \text{det-ladder } \{8, 14, 20\}$

External-review residual map [E]/[C] ([verification/v182_reviewer_residual_map.py])

A careful external review (2026-06-14) raised four concerns at the [C]/[O] seams: **(A)** the mapping $2D \to 4D$ ("why does the dynamics read the masses as the Plücker norm 11?"), **(B)** the abstractness of UV gravity, **(C)** the meta-look-elsewhere of the grammar choice, and **(D)** the Koide Möbius flow (non-integer flow time $t \sim 2.84$, missing QFT generator). None adds an *independent* open gap. **A** [E]/[C]: the readout integer is exact ($11 = \sum_{k\le 2}\binom{4}{k} = \dim S^+ - g_{\rm car}$, the QBL boundary count, v174) and the $2D \to 4D$ step is the holographic reduction, so $A = \texttt{QGEO.SYM.01} + F_{\rm transfer}$. **B** [E]: "gravity is the geometric readout of the boundary" *is* QGEO.SYM.01. **D** [E]/[C]: $\operatorname{Aut}(\mathbb{P}) = \mathrm{PSL}(2, \mathbb{C})$ is the Möbius group and the cusp flow lives on the deck $\mathbb{P} \setminus \mu_4$, so the Möbius *nature* is forced by the geometry; the non-integer $t \sim 2.84$ only locates the physical point and the missing generator *is* $F_{\rm transfer}$, so $D = \texttt{QGEO.SYM.01} + F_{\rm transfer}$. So A, B, D collapse onto the *two already-named* residuals; only **C** is genuinely *experiment-only*: the grammar is frozen (v84), minimal and over-determined (the anchor $a = (1, 1, 2)$ lands in ≥ 6 mutually-independent, not-selected-for structures), a look-elsewhere-resistant signature whose residual — correctly — falls only to out-of-sample data (JUNO, CMB-S4), never to argument. The four concerns are a unification of the residual *structure*, not a closure of the underlying premises, which stay [O]/[C].

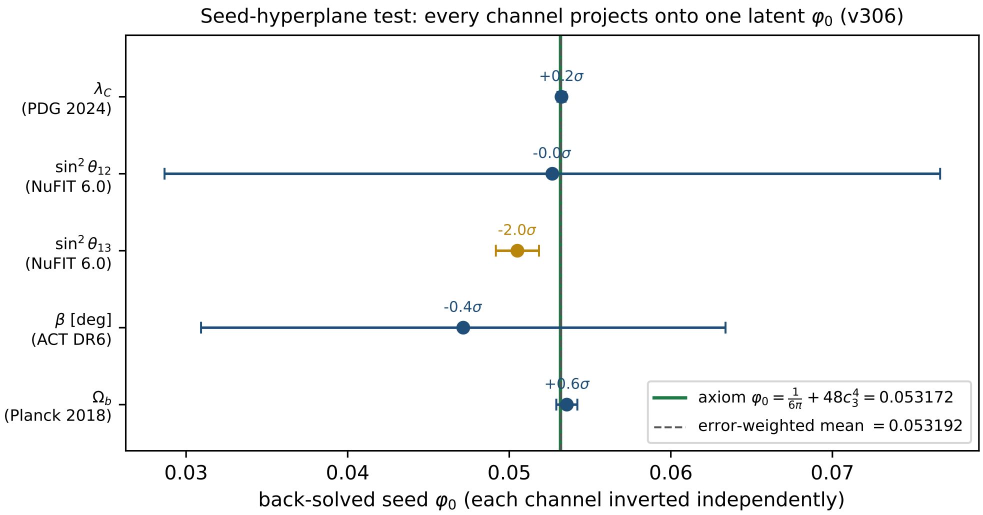

Figure 1: The seed-hyperplane test ([`verification/v306_seed_crossval.py`]): each scorecard observable is *independently* inverted to the latent seed φ_0 (with its measurement error). All five project onto the one axiom value $\varphi_0 = \frac{1}{6\pi} + 48c_3^4 = 0.053172$ (green), and their error-weighted mean reproduces it to 0.04% — one real seed, not five fits. The honest pressure point is $\sin^2\theta_{13}$ ($\sim 2\sigma$ low, a *separate* carrier-trace channel, v268); a data↔formula shuffle explodes the cross-validation χ^2 by $> 100\times$. [**E**].